AeroCorridor-6G
WHERE THE SKY LEARNS TO FLY

System Architecture & Technical Concept Note

1. Introduction

AeroCorridor-6G is a next-generation autonomous aerial traffic management platform designed for the emerging era of unmanned drones, defence fleets, and urban air mobility. As drone activity scales globally, existing navigation systems—which rely on GPS, cameras, or predefined static routes—are insufficient for safe, predictable, and large-scale autonomous operations.

Our system introduces a 6G-native approach where communication signals are also used for perception. This allows the network to sense, interpret, coordinate, and govern movement across multiple drones and future air-taxis—without requiring external sensors or human intervention.

2. Core Technical Principle: ISAC

At the heart of AeroCorridor-6G is Integrated Sensing and Communication (ISAC) — a foundational capability envisioned for 6G.

Instead of using separate hardware for communication and sensing, ISAC enables:

- Simultaneous environment sensing and data transmission
- RF-based spatial awareness in real time
- GPS-independent perception in denied or jammed conditions

This is critical for future deployments in dense urban environments, disaster zones, defence airspace, and low-visibility atmospheric conditions.

3. System Architecture Overview

AeroCorridor-6G consists of four primary layers:

Layer	Function
ISAC Signal Layer	Captures Doppler, phase, range, and reflection data while transmitting telemetry
Sensing & Fusion Engine	Converts raw RF patterns into a shared 3D situational awareness map
AI-Native Control & Routing Layer	Uses reinforcement learning to predict congestion, optimize flight lanes, and prevent collisions
Secure Execution & Compliance Layer	Handles encrypted command execution, identity verification, fallback modes, and PQC resilience

4. Multi-Node Coordination

Unlike isolated drone autonomy, AeroCorridor-6G synchronizes intelligence across multiple air units. Every drone becomes a node in a distributed sensing network, increasing:

- Accuracy
- Redundancy
- Resilience
- Trust

This transforms the airspace into a self-aware, self-managing system.

5. Prototype Scope

The first functional prototype demonstrates:

- **Simultaneous sensing & data transmission**
- **Motion and obstacle detection via simulated RF signals**
- **Three-drone sensing fusion using digital-twin environments**
- **AI-based route optimization and lane assignment**

The goal is to prove feasibility of **ISAC-driven autonomy without GPS or external perception.**

6. Alignment With 6G Vision

AeroCorridor-6G directly aligns with:

6G Focus Area	Alignment
ISAC	Core sensing & control method
Native AI	Routing, prediction, and decision-making
NTN & Beyond-5G Mobility	Multi-layer connectivity (ground + satellite)
Secure & Trusted Autonomy	PQC identity-bound swarm control
Digital Infrastructure for Smart Cities	Scalable public safety & logistics automation

7. Why This Matters

Autonomous mobility will not scale unless the sky becomes predictable and safe.
AeroCorridor-6G provides the missing layer:

A shared, intelligent, self-governed aerial infrastructure — not just a drone control system.

8. Expected Outcomes

✔ Prototype demonstration
✔ Patent-eligible sensing + routing models
✔ Telecom + regulator integration roadmap
✔ Deployment-ready blueprint for urban or defence corridors

9. Technical Alignment & Patentable Differentiators

📡 Spectrum & Connectivity Framework

AeroCorridor-6G is designed to operate across multiple connectivity modes, including:

- **5G/5G+ terrestrial URLLC slices**
- **NTN (Non-Terrestrial Networks)** for beyond-LOS drone paths
- **mmWave / Sub-THz sensing bands** for high-resolution ISAC mapping
- **Edge compute mesh** for ultra-low latency decision execution

This ensures interoperability across **urban, remote, and defence-restricted airspace.**

📜 Standards Compatibility (3GPP Alignment)

The system is designed in alignment with emerging standards for:

Standard Element	Alignment
3GPP Rel-19 ISAC Framework	✔ Used for sensing + comm dual-function signals
Rel-20 NTN Mobility	✔ Enables satellite-supported autonomous flight
AI-Native Network Management	✔ RL routing + multi-node situational fusion
UTM/UAM Air Traffic Compliance	✔ Integration path to Digital Sky + FAA/EASA models

AeroCorridor-6G is designed to ensure that as aerial mobility evolves, it evolves with **intelligence, safety, and trust.** It represents a step toward a future where the sky itself becomes a network — able to sense, understand, and coordinate everything that flies and includes **Bharat 6G Mission alignmen**t and **Telecom + OEM integration.**

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AeroCorridor-6G — Where the sky learns to fly.